

Choose Inputs and Outputs

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Find a Real Problem

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Verify Important Outputs

Without opening the old lesson, explain one key idea from Verify Important Outputs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Privacy by Choosing Less

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Run the baseline on three requests for Requirements and a simple baseline. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Requirements and a simple baseline. Proposed explanation: “Quiet required No excludes quiet activities”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Build a baseline for a different catalogue

A paper room recommender uses R1 first, capacity 6, quiet Yes; then R2, capacity 10, quiet No. Valid requests give a whole group size from 1 to 12 and Quiet required Yes or No. A room fits when capacity is at least group size and the quiet condition is met; choose the first fitting room. Record outputs and reasons for 6 people Yes, 8 people No and 8 people Yes. Explain how a manual baseline can be compared fairly with a prototype. No real booking is allowed.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Design a paper/plastic sorter that receives an object description. Include a mixed paper-and-plastic cup. Specify inputs, outputs and the uncertain-case action. Is the owner's name useful?

6. Solve the given case

A sorter suggests paper or plastic trays using material observations.

Name an input, output and unnecessary input.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A paper sorter must suggest pencil tray or book shelf. Choose one needed input and state the output. What should happen if the picture is blocked?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 30 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Freeze the sorter's v1 input list, output contract, and not-sure behavior.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Design a paper/plastic sorter that receives an object description. Include a mixed paper-and-plastic cup. Specify inputs, outputs and the uncertain-case action. Is the owner's name useful?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Make a requirement checkable

Replace “suggest a good activity” with the time condition, quiet condition and tie rule used in this project. Explain why Quiet required No permits B1 as well as non-quiet activities when their times fit.

My reasoning and evidence

13. W2 Run the baseline on three requests

Using the catalogue, record the baseline output and reason for 10 minutes with quiet required, 8 minutes with quiet not required, and 7 minutes with quiet not required. Identify which no-match result is justified and why.

My reasoning and evidence

14. W3 Separate invalid input from no matching option

Compare a request with blank minutes and Quiet required Yes to a request with 0 minutes and Quiet required Yes. What should the baseline record for each? Explain why these are different outcomes even though neither produces an activity suggestion.

My reasoning and evidence

15. W4 Build a baseline for a different catalogue

A paper room recommender uses R1 first, capacity 6, quiet Yes; then R2, capacity 10, quiet No. Valid requests give a whole group size from 1 to 12 and Quiet required Yes or No. A room fits when capacity is at least group size and the quiet condition is met; choose the first fitting room. Record outputs and reasons for 6 people Yes, 8 people No and 8 people Yes. Explain how a manual baseline can be compared fairly with a prototype. No real booking is allowed.

My reasoning and evidence

16. W5 Challenge an unfair comparison

Team A tests a prototype on five easy requests and the manual baseline on five different requests, then reports only the prototype's successful outputs. Identify two changes needed for a fair comparison. If both methods then meet all requirements, what can and cannot be concluded about compatibility?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Choose Inputs and Outputs

A fictional classroom chooses a storage bin for model parts.

Catalogue order: B1 capacity 5, B2 capacity 9. Valid request is a whole part count 1–12.

Return first bin with capacity at least the count; none if neither fits; clarify invalid input.

Choose Inputs and Outputs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Validate request

Count 6 is whole and within 1-12.



2. Check first bin

B1 capacity 5 is below 6; reject it.



3. Check second bin

B2 capacity 9 is at least 6; accept it.



4. Baseline output

Report B2 and stop; save the comparison evidence.

CHANGE ONE THING

Change request 6 to 5.

Explain the changed result using the matching lesson.

Trace requests 10 and blank. Why should they not receive the same explanation?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Document sorter schema, catalogue order, acceptance rule and all output types; use a manual trace as the comparison baseline.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

Document sorter schema, catalogue order, acceptance rule and all output types; use a manual trace as the comparison baseline.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional classroom chooses a storage bin for model parts.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report B2 and stop; save the comparison evidence.

1. Baseline output
2. Check first bin
3. Validate request
4. Check second bin

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Baseline output Report B2 and stop; save the comparison evidence. The baseline is an ordered catalogue scan, not a preference ranking. It makes expectations testable before a prototype is built. Count 10 is valid but has no matching bin; that differs from a blank count, which fails the input contract. Both situations need explicit output behavior.

Practice 2: explain the deciding evidence

Changed-case practice: Change request 6 to 5. Which conclusion is supported by the supplied facts?

1. The baseline returns B1.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The baseline returns B1. Its capacity equals the request, and at least includes equality.

Repair a missing building block

Review Find a Real Problem (ai-1/29). Goal: Describe a specific user difficulty using supplied evidence without inventing its cause. Short practice: Write a one-sentence project goal and say what evidence would count as a failure.. Return to this week and explain what you changed.

Review Verify Important Outputs (ai-1/27). Goal: Separate an answer into claims and connect each claim to relevant evidence. Short practice: Correct the draft and identify which part needs another source rather than a calculation.. Return to this week and explain what you changed.

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields. Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
